

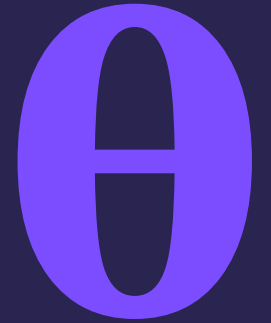
IB MATH AA SL

Lesson 17

Radians, Arcs & Sectors

Trigonometry · 60-minute lesson

Syllabus SL 3.4 · Property of Lynda Badmus Education

A large, stylized Greek letter theta (θ) is positioned in the upper right quadrant of the slide. It is rendered in a light blue color, matching the header bar. The background of the slide features two large, overlapping circles in shades of blue and purple, creating a modern, geometric aesthetic.

Do Now — recall

ON YOUR WHITEBOARD

1. How many degrees in a full turn?
2. What fraction of a circle is 90° ?
3. Circumference of a circle radius r ?

Answers (click to reveal)

1. 360°
2. one quarter
3. $2\pi r$

What we're learning today

LEARNING OBJECTIVES

- 1 Convert**
between degrees and radians.
- 2 Find**
the length of an arc.
- 3 Find**
the area of a sector.
- 4 Apply**
these to real contexts.

SUCCESS CRITERIA — I CAN...

- ✓
convert degrees $\leftrightarrow$ radians
- ✓
use $l = r\theta$ for arc length
- ✓
use $A = \frac{1}{2}r^2\theta$ for sector area
- ✓
keep θ in radians for these formulas

Radians

DEGREES ↔ RADIANS

$$\pi \text{ rad} = 180^\circ$$

To radians

$\times \pi/180$

To degrees

$\times 180/\pi$

Why

one radian = the angle that an arc of length r subtends

COMMON ANGLES

90°

$\pi/2$

180°

π

60°

$\pi/3$

45°

$\pi/4$

Convert an angle

QUESTION

Convert 120° to radians,
giving an exact answer.

SOLUTION

Multiply by $\pi/180$:

$$120 \times \frac{\pi}{180}$$

Simplify the fraction ($120/180 = 2/3$):

$$= \frac{2\pi}{3}$$

Your turn — convert

ANSWER IN YOUR BOOK

1. Convert 90° to radians.
2. Convert $\pi/6$ to degrees.
3. Convert 270° to radians.

Answers (click to reveal)

1. $\pi/2$
2. 30°
3. $3\pi/2$

Arcs & sectors

WITH θ IN RADIANS

$$l = r \theta$$

$$A = \frac{1}{2} r^2 \theta$$

$$l = r\theta$$

arc length

$$A = \frac{1}{2} r^2 \theta$$

sector area

Important

θ must be in RADIANS

WHERE FROM

A full circle: $\theta = 2\pi$.

Arc

$$l = r(2\pi) = 2\pi r \quad \checkmark$$

Area

$$A = \frac{1}{2} r^2 (2\pi) = \pi r^2 \quad \checkmark$$

Arc length & area

QUESTION

A sector has radius 6 cm and angle 0.5 rad.

Find the arc length and the area.

SOLUTION

Arc length $l = r\theta$:

$$l = 6 \times 0.5 = 3 \text{ cm}$$

Sector area $A = \frac{1}{2}r^2\theta$:

$$A = \frac{1}{2}(6)^2(0.5) = 9 \text{ cm}^2$$

Exam-style question

A sector of a circle has radius 10 cm and angle 1.2 radians.

- (a) Find the arc length. [2]
- (b) Find the area of the sector. [2]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $l = r\theta = 10 \times 1.2 = 12 \text{ cm}$ (M1)(A1)

(b) $A = \frac{1}{2}r^2\theta = \frac{1}{2}(100)(1.2) = 60 \text{ cm}^2$ (M1)(A1)

Common mistakes



Mode

Set your GDC to RADIANS for these formulas.



Degrees in $l = r\theta$

θ must be in radians, not degrees.



Half

Sector area is $\frac{1}{2}r^2\theta$ — don't forget the $\frac{1}{2}$.



Convert carefully

$\times \pi/180$ for degrees $\rightarrow$ radians.

Mixed practice

QUESTIONS

1. Convert 150° to radians.
2. Arc length: $r = 8$, $\theta = 0.75$ rad.
3. Sector area: $r = 5$, $\theta = 1.6$ rad.
4. Convert $2\pi/5$ to degrees.

Answers (click to reveal)

1. $5\pi/6$
2. 6 cm
3. 20 cm^2
4. 72°

Plenary & exit ticket

KEY FORMULAS

$$\pi = 180^{\circ}$$

$$l = r\theta, \quad A = \frac{1}{2}r^2\theta$$

EXIT TICKET — before you leave

1. Convert 60° to radians.
2. Arc length: $r = 4$, $\theta = 2$ rad.
3. Sector area: $r = 3$, $\theta = 2$ rad.

Answers (click to reveal)

1. $\pi/3$
2. 8
3. 9

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Radians, arcs & sectors.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 18

The unit circle & exact values of sin, cos and tan.

$$\sin^2 \theta + \cos^2 \theta = 1$$